

Git & GitHub

Interview Questions & Answers

Part 1 — Git Questions (Q1–Q25)

Q1. What is Git?

Answer

Git is a distributed version control system used to track changes in source code and collaborate efficiently with developers.

Q2. Why is Git used?

Answer

Git is used to manage code versions, collaborate with teams, maintain history, and recover previous versions of projects.

Q3. What is a repository?

Answer

A repository (repo) is a storage location that contains project files, folders, and complete version history.

Q4. Difference between Git and GitHub?

Answer

Git	GitHub
Version control tool	Cloud-based hosting platform
Works locally	Provides online repository hosting
Uses command line	Provides GUI and web interface

Q5. What is version control?

Answer

Version control is a system that tracks changes in files and allows developers to manage different versions of code.

Q6. What is git init?

Answer

git init initializes a new Git repository. It creates a hidden .git directory that stores Git information.

Command:

```
git init
```

Q7. What is git clone?

Answer

git clone copies an existing remote Git repository to your local machine.

Command:

```
git clone <repository_url>
```

Q8. What is git add?

Answer

git add stages new or modified files so they can be included in the next commit.

Command:

```
git add .
```

Q9. What is git commit?

Answer

git commit saves staged changes permanently in the repository history with a descriptive message.

Command:

```
git commit -m "message"
```

Q10. What is git status?

Answer

git status displays the current state of the working directory and staging area.

Command:

```
git status
```

Q11. What is git push?

Answer

git push uploads local commits to a remote repository.

Command:

```
git push <remote> <branch>
```

Q12. What is git pull?

Answer

git pull downloads changes from a remote repository and merges them into the current branch.

Command:

```
git pull <remote>
```

Q13. What is git fetch?

Answer

git fetch downloads changes from a remote repository without automatically merging them.

Command:

```
git fetch <remote>
```

Q14. Difference between Git Pull and Git Fetch?

Answer

Git Pull	Git Fetch
Downloads and merges changes	Only downloads changes
Updates current branch automatically	Requires manual merge

Q15. What is git log?

Answer

git log shows the commit history of a repository.

Command:

```
git log
```

Q16. What is a branch?

Answer

A branch is a separate development line that allows developers to work on features without affecting the main codebase.

Command:

```
git branch <branch_name>
```

Q17. What is Merge?

Answer

Merge is the process of combining changes from one branch into another branch.

Command:

```
git merge <branch_name>
```

Q18. What is a Merge Conflict?

Answer

A merge conflict occurs when Git cannot automatically combine changes because the same part of a file was modified differently in two branches.

Solution:

Resolve conflicts manually and commit the changes.

Q19. What is HEAD in Git?

Answer

HEAD is a pointer that represents the current branch or commit you are working on.

Q20. What is the Staging Area?

Answer

The staging area is an intermediate area where changes are prepared before committing them to the repository.

Command:

```
git add <file>
```

Q21. What is git stash?

Answer

git stash temporarily saves uncommitted changes so you can switch branches or perform other operations.

Command:

```
git stash
```

Q22. What is git reset?

Answer

git reset moves the HEAD pointer backward and can undo commits or changes.

Commands:

```
git reset --soft <commit_hash>  
git reset --hard <commit_hash>
```

Q23. What is git revert?

Answer

git revert creates a new commit that reverses the changes made by a previous commit.

Q24. Difference between Git Reset and Git Revert?

Answer

Git Reset	Git Revert
Removes commits from history	Creates a new reverse commit
Can change history	Keeps history safe
Mostly used locally	Used for shared repositories

Q25. What is a Remote Repository?

Answer

A remote repository is a Git repository hosted on a server such as GitHub, GitLab, or Bitbucket that allows collaboration.

Part 2 — GitHub Questions (Q26–Q50)

Q26. What is GitHub?

Answer

GitHub is a cloud-based platform that hosts Git repositories and provides collaboration features such as code sharing, pull requests, issue tracking, and project management.

Q27. What is the difference between Local Repository and Remote Repository?

Answer

Local Repository	Remote Repository
Stored on your computer	Stored on platforms like GitHub
Used for personal development	Used for collaboration
Works without internet	Requires network connection for synchronization

Q28. What is GitHub Repository?

Answer

A GitHub repository is an online storage space where project files, source code, documentation, and version history are stored.

Q29. What is Fork in GitHub?

Answer

A fork creates a personal copy of another user's repository under your GitHub account. It allows you to modify the project without affecting the original repository.

Q30. What is a Pull Request (PR)?

Answer

A Pull Request is a request to merge changes from one branch or repository into another. It allows code review and discussion before merging.

Q31. What is GitHub Issue?

Answer

GitHub Issues are used to track bugs, feature requests, improvements, and tasks within a repository.

Q32. What is GitHub Actions?

Answer

GitHub Actions is a CI/CD automation tool that allows developers to automate workflows such as testing, building, and deploying applications.

Example:

- Run tests automatically after every push
- Deploy applications automatically

Q33. What is .gitignore file?

Answer

.gitignore is a file that specifies which files or folders Git should ignore and not track.

Examples:

```
node_modules/  
.env  
__pycache__/
```

Q34. Why do we use .gitignore?

Answer

We use .gitignore to prevent unnecessary or sensitive files from being uploaded to the repository.

Examples:

- Password files
- API keys
- Temporary files
- Dependencies

Q35. What is GitHub Branch Protection?

Answer

Branch protection rules prevent unauthorized changes to important branches like main or master.

Features:

- Require code review
- Require successful tests
- Prevent force pushes

Q36. What is GitHub Actions Workflow?

Answer

A workflow is an automated process defined using a YAML file inside:

```
.github/workflows/
```

Example:

```
name: Python Test

on:
  push:

jobs:
  test:
    runs-on: ubuntu-latest
```

Q37. What is Continuous Integration (CI)?

Answer

Continuous Integration is a development practice where developers frequently merge code changes and automatically test them.

Benefits:

- Finds errors early
- Improves code quality
- Reduces integration problems

Q38. What is Continuous Deployment (CD)?

Answer

Continuous Deployment automatically releases tested code changes to production.

Example:

```
Code Push → Test → Build → Deploy
```

Q39. What is Git Tag?

Answer

A Git tag is used to mark specific points in Git history, usually for software releases.

Example:

```
git tag v1.0
```

Q40. What is GitHub Release?

Answer

GitHub Release allows developers to publish a specific version of software with release notes and downloadable files.

Example:

```
v1.0.0  
v2.0.0
```

Q41. What is GitHub README.md?

Answer

README.md is a documentation file that explains project details, installation steps, usage, and features.

Q42. What is Markdown?

Answer

Markdown is a lightweight markup language used to format documentation files like README.md.

Example:

```
# Project Title  
  
## Installation
```

Q43. What is GitHub Star?

Answer

A GitHub star is a way to bookmark and show appreciation for useful repositories.

Q44. What is GitHub Watch?

Answer

GitHub Watch allows users to receive notifications about activities in a repository.

Q45. What is GitHub Organization?

Answer

A GitHub Organization is a shared account used by teams and companies to manage repositories and permissions.

Q46. What is GitHub Collaborator?

Answer

A collaborator is a person who is given permission to access and contribute to a repository.

Q47. What is GitHub Security?

Answer

GitHub Security provides tools to identify and fix security vulnerabilities.

Features:

- Dependabot
- Secret scanning
- Code scanning

Q48. What is GitHub Pages?

Answer

GitHub Pages is a service that hosts static websites directly from a GitHub repository.

Example uses:

- Portfolio websites
- Documentation websites
- Project demos

Q49. What is GitHub Copilot?

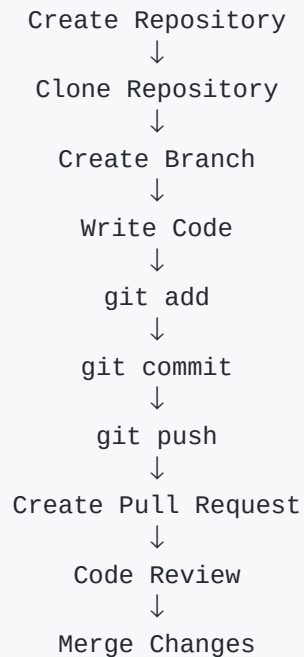
Answer

GitHub Copilot is an AI-powered coding assistant that helps developers write, understand, and improve code.

Q50. Explain a basic Git workflow.

Answer

A basic Git workflow:



Commands:

```
git clone <url>

git checkout -b feature

git add .

git commit -m "Add feature"

git push origin feature
```

End of Document — All 50 Git & GitHub interview questions with corrected spelling and grammar.